Lecture 5. Nonparametric Estimation of Basic Quantities for Right-Censored Data (Part 1)

Kaplan–Meier and Nelson–Aalen Estimators

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1 Setting

X is a continuous failure time with survival function $S(t)$ and cumulative hazard $H(t)$. X is subject to **non-informative right censoring**, so we observe

$$\{(T_i, \delta_i),\ i = 1, \ldots, n\}, \qquad T_i = \min(X_i, C_i), \quad \delta_i = I(X_i \le C_i).$$

Goal: estimate $S(t)$ and $H(t)$ without assuming a parametric family. We build up the estimator in four steps: discrete/continuous $\times$ without/with censoring.

2 Building Up the Estimator

2.1 Discrete X without censoring

Let X have M support points $0 < \tilde{x}_1 < \cdots < \tilde{x}_M < \infty$. With $\tilde{x}_0 = 0$, $S(\tilde{x}_0) = 1$, the product-limit representation (Lecture 2) is

$$S(x) = P(X > x) = \prod_{\tilde{x}_j \le x} \frac{S(\tilde{x}_j)}{S(\tilde{x}_{j-1})} = \prod_{\tilde{x}_j \le x} P(X > \tilde{x}_j \mid X > \tilde{x}_{j-1})$$

$$= \prod_{\tilde{x}_j \le x} P(X > \tilde{x}_j \mid X \ge \tilde{x}_j) = \prod_{\tilde{x}_j \le x} [1 - h(\tilde{x}_j)], \qquad h(\tilde{x}_j) = P(X = \tilde{x}_j \mid X \ge \tilde{x}_j).$$

(The third equality holds because there is no mass strictly between $\tilde{x}_{j-1}$ and $\tilde{x}_j$.) With i.i.d. $X_1, \ldots, X_n$, estimate

$$\hat{h}(\tilde{x}_j) = \frac{\#\{X_i = \tilde{x}_j\}}{\#\{X_i \ge \tilde{x}_j\}} = \frac{\#\text{ deaths at } \tilde{x}_j}{\#\text{ deaths at } \tilde{x}_j, \tilde{x}_{j+1}, \ldots, \tilde{x}_M}, \qquad \hat{S}(x) = \prod_{\tilde{x}_j \le x} [1 - \hat{h}(\tilde{x}_j)].$$

$\hat{h}$ and $\hat{S}$ are the MLEs of h and S.

2.2 Discrete X with right censoring

With $T_i = \min(X_i, C_i)$, $\delta_i = I(T_i = X_i)$ and $X_i \perp\!\!\!\perp C_i$:

$$\begin{aligned} P(T_i = \tilde{x}_j, \delta_i = 1 \mid T_i \ge \tilde{x}_j) &= \frac{P(T_i = \tilde{x}_j, \delta_i = 1)}{P(T_i \ge \tilde{x}_j)} = \frac{P(X_i = \tilde{x}_j,\ C_i \ge \tilde{x}_j)}{P(X_i \ge \tilde{x}_j,\ C_i \ge \tilde{x}_j)} \\ &= \frac{P(X_i = \tilde{x}_j)P(C_i \ge \tilde{x}_j)}{P(X_i \ge \tilde{x}_j)P(C_i \ge \tilde{x}_j)} = \frac{P(X_i = \tilde{x}_j)}{P(X_i \ge \tilde{x}_j)} = h(\tilde{x}_j). \end{aligned}$$

So the hazard is identifiable from the *observed* data, which suggests

$$\hat{h}(\tilde{x}_j) = \frac{\#\{T_i = \tilde{x}_j, \delta_i = 1\}}{\#\{T_i \ge \tilde{x}_j\}}, \qquad \hat{S}(x) = \prod_{\tilde{x}_j \le x} \Big[1 - \frac{\#\{T_i = \tilde{x}_j, \delta_i = 1\}}{\#\{T_i \ge \tilde{x}_j\}}\Big].$$

Derivation: $\hat{h}$ is the MLE under censoring

Under independent censoring, the likelihood in the hazard parameters factorizes over support points. With d_j deaths among Y_j at risk at $\tilde{x}_j$, each subject at risk either dies at $\tilde{x}_j$ (prob. h_j) or survives past it (prob. $1 - h_j$); censored subjects contribute survival up to their censoring time. Thus

$$L(h_1, \ldots, h_M) \propto \prod_j h_j^{d_j}(1 - h_j)^{Y_j - d_j}, \qquad \frac{\partial \log L}{\partial h_j} = \frac{d_j}{h_j} - \frac{Y_j - d_j}{1 - h_j} = 0 \ \Rightarrow\ \hat{h}_j = \frac{d_j}{Y_j}.$$

By invariance, $\hat{S}(x) = \prod_{\tilde{x}_j \le x}(1 - \hat{h}_j)$ is the MLE of S.

2.3 Continuous X

Without censoring, the survival function is smooth and can be estimated by the complement of the empirical distribution function:

$$\hat{S}(t) = 1 - \mathbb{F}_n(t) = 1 - \frac{1}{n}\sum_{i=1}^{n} I(X_i \le t).$$

With right censoring we must modify this. Two questions guide the construction: *Should the estimator jump at censoring times?* (No — no event is observed there.) *What is the contribution of a censored observation?* (It counts in the risk set Y_j for every death time up to its censoring time, then leaves.)

3 The Kaplan–Meier (Product-Limit) Estimator

Motivation. With censored data the empirical survival function cannot be computed, since we do not know when censored subjects fail. Kaplan–Meier writes survival as a product of conditional survival probabilities over the observed failure times, each estimated from those still at risk.

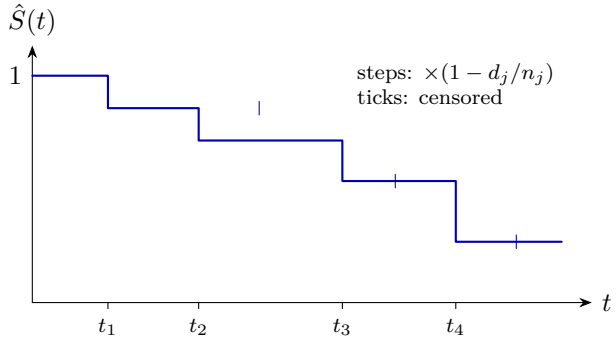


Figure. *The Kaplan–Meier estimate is a step function that drops only at observed failure times t_j, by the factor $1 - d_j/n_j$. Censored observations (ticks) cause no drop but leave the risk set, which makes later drops larger.*

Definition 5.1: Notation

- $t_1 < t_2 < \cdots < t_D$: the D distinct death times;
- $d_j = \sum_{i=1}^{n} I(T_i = t_j, \delta_i = 1)$: number of deaths at t_j;
- $Y_j = \sum_{i=1}^{n} I(T_i \geq t_j)$: number at risk (alive and under observation) just before t_j.

Theorem 5.2: Kaplan–Meier estimator

$$\hat{S}(t) = \begin{cases} 1, & t < t_1, \\ \displaystyle\prod_{t_j \leq t} \left(1 - \frac{d_j}{Y_j}\right), & t_1 \leq t. \end{cases}$$

Remark 5.3: Ties between deaths and censorings

If a death and a censoring occur at the same recorded time, the convention is that the censoring happens just *after* the death, so the censored subject is still counted in Y_j (Y_j uses $T_i \geq t_j$).

Theorem 5.4: Greenwood's variance estimator

$$\hat{V}[\hat{S}(t)] = \hat{S}(t)^2 \sum_{t_j \leq t} \frac{d_j}{Y_j(Y_j - d_j)}.$$

Derivation: delta-method derivation (Cox & Oakes, 1984)

Let $\hat{h}_j = d_j/Y_j$. Treating $d_j \mid Y_j \sim \text{Bin}(Y_j, h_j)$ and the $\hat{h}_j$ as approximately uncorrelated

across j,

$$\mathrm{Var}(\hat{h}_j) \approx \frac{\hat{h}_j(1-\hat{h}_j)}{Y_j}.$$

Since $\ln \hat{S}(t) = \sum_{t_j \le t} \ln(1-\hat{h}_j)$ and, by the delta method with $g(u) = \ln(1-u)$, $g'(u) = -1/(1-u)$:

$$\begin{aligned}
\mathrm{Var}\{\ln \hat{S}(t)\} &\approx \sum_{t_j \le t} \mathrm{Var}\{\ln(1-\hat{h}_j)\} \approx \sum_{t_j \le t} \Big(\frac{1}{1-\hat{h}_j}\Big)^2 \mathrm{Var}(\hat{h}_j) \\
&\approx \sum_{t_j \le t} \Big(\frac{1}{1-\hat{h}_j}\Big)^2 \frac{\hat{h}_j(1-\hat{h}_j)}{Y_j} = \sum_{t_j \le t} \frac{\hat{h}_j}{Y_j(1-\hat{h}_j)} = \sum_{t_j \le t} \frac{d_j/Y_j}{Y_j(Y_j-d_j)/Y_j} \\
&= \sum_{t_j \le t} \frac{d_j}{Y_j(Y_j-d_j)}.
\end{aligned}$$

A second delta-method step with $\hat{S} = \exp(\ln \hat{S})$ gives $\hat{V}[\hat{S}(t)] = \hat{S}(t)^2 \,\mathrm{Var}\{\ln \hat{S}(t)\}$.

4 The Nelson–Aalen Estimator

Motivation. Estimating the cumulative hazard by summing the observed failure rates d_i/n_i gives the Nelson–Aalen estimator, which has better small-sample properties and is the natural input for graphical model checks.

Theorem 5.5: Nelson–Aalen estimator and Aalen's variance

$$\tilde{H}(t) = \begin{cases} 0, & t < t_1, \\ \displaystyle\sum_{t_j \le t} \frac{d_j}{Y_j}, & t_1 \le t, \end{cases} \qquad \sigma_H^2(t) = \sum_{t_j \le t} \frac{d_j}{Y_j^2}.$$

Remark 5.6: Intuition

$H(t) = \int_0^t h(s)\,\mathrm{d}s$ accumulates the hazard; at each death time the hazard increment is estimated by $\hat{h}_j = d_j/Y_j$. Aalen's variance treats d_j as approximately Poisson with mean $Y_j h_j$, so $\mathrm{Var}(d_j/Y_j) \approx Y_j h_j / Y_j^2 \approx d_j/Y_j^2$.

4.1 Alternative estimators

- An alternative estimator of H: $\hat{H}(t) = -\log \hat{S}(t)$.
- An alternative estimator of S: $\tilde{S}(t) = \exp\{-\tilde{H}(t)\}$ (the Breslow / Fleming–Harrington estimator).
- Since $1 - x \le e^{-x}$, each KM factor $1 - d_j/Y_j \le e^{-d_j/Y_j}$, so $\hat{S}(t) \le \tilde{S}(t)$ and equivalently $\tilde{H}(t) \le \hat{H}(t)$.
- $\hat{S}(t)$ is called the "product-limit" estimator: as $n \uparrow \infty$, $D \uparrow \infty$ and the number of terms in the product $\to \infty$ for continuous X. For finite n, $D < \infty$; in the limit the product runs over the whole support of X (a *product integral*).

5 Worked Example (Small Sample)

Example 5.7: $n = 5$

Continuous X subject to right censoring, $T = \min(X, C)$, $\delta = I(X \le C)$:

T_i	0.5	1	0.75	0.25	0.75
δ_i	1	1	1	0	0

Ordered: 0.25^+, 0.5, 0.75, 0.75^+, 1. At $t = 0.75$ the censored subject is still at risk (tie convention).

t_j	d_j	Y_j	$1 - d_j/Y_j$	$\hat{S}(t_j)$	d_j/Y_j	$\tilde{H}(t_j)$	$\tilde{S}(t_j) = e^{-\tilde{H}}$
0.5	1	4	3/4	0.750	0.250	0.250	0.779
0.75	1	3	2/3	0.500	0.333	0.583	0.558
1	1	1	0	0.000	1.000	1.583	0.205

Greenwood: $\hat{V}[\hat{S}(0.5)] = 0.75^2 \cdot \frac{1}{4 \cdot 3} = 0.0469$, $\hat{V}[\hat{S}(0.75)] = 0.5^2(\frac{1}{12} + \frac{1}{3 \cdot 2}) = 0.0625$; at $t = 1$, $Y_j - d_j = 0$ and Greenwood is undefined (the KM drops to 0).

6 Remarks

- $\hat{S}$, $\tilde{H}$, $\hat{H}$ are right-continuous step functions even though S and H may be continuous.
- $\hat{S}$ and $\tilde{S}$ are consistent for $S(t)$ for all $t \le \tau$ such that $P(T \ge \tau) > 0$, and are asymptotically equivalent.
- $\hat{S}$ and $\tilde{H}$ are only defined for $t \le \max(T_i) = t_{\max}$. If the largest observation is censored, $\hat{S}$ does not reach 0. Some extensions for $t > t_{\max}$:
  - Efron (1967): $\hat{S}(t) = 0$;
  - Gill (1980): $\hat{S}(t) = \hat{S}(t_{\max})$;
  - Brown, Hollander & Kowar (1974): $\hat{S}(t) = \exp\{t \log[\hat{S}(t_{\max})]/t_{\max}\}$ (an exponential tail).

7 The 6-MP Leukaemia Trial

Data. 42 children with acute leukaemia in complete or partial remission were matched in pairs by remission status and randomized within the pair to 6-mercaptopurine (6-MP) or placebo maintenance therapy, then followed until relapse or end of study. Times to relapse (weeks) for the 21 **6-MP** patients ($^+$ = censored):

$$6,\ 6,\ 6,\ 6^+,\ 7,\ 9^+,\ 10,\ 10^+,\ 11^+,\ 13,\ 16,\ 17^+,\ 19^+,\ 20^+,\ 22,\ 23,\ 25^+,\ 32^+,\ 32^+,\ 34^+,\ 35^+.$$

Placebo times (all observed): $1, 1, 2, 2, 3, 4, 4, 5, 5, 8, 8, 8, 8, 11, 11, 12, 12, 15, 17, 22, 23$.

Example 5.8: Kaplan–Meier estimate for the 6-MP group

t_j	d_j	Y_j	$\hat{S}(t) = \prod(1 - d_j/Y_j)$	$\sum \frac{d_j}{Y_j(Y_j - d_j)}$	$\hat{S}(t)^2 \sum \cdots$
6	3	21	$[1 - \frac{3}{21}] = 0.857$	$\frac{3}{21\times 18} = 0.0079$	0.0058
7	1	17	$0.857(1 - \frac{1}{17}) = 0.807$	$0.0079 + \frac{1}{17\times 16} = 0.0116$	0.0076
10	1	15	$0.807(1 - \frac{1}{15}) = 0.753$	$0.0116 + \frac{1}{15\times 14} = 0.0164$	0.0093
13	1	12	$0.753(1 - \frac{1}{12}) = 0.690$	$0.0164 + \frac{1}{12\times 11} = 0.0240$	0.0114
16	1	11	$0.690(1 - \frac{1}{11}) = 0.628$	$0.0240 + \frac{1}{11\times 10} = 0.0330$	0.0130
22	1	7	$0.628(1 - \frac{1}{7}) = 0.538$	$0.0330 + \frac{1}{7\times 6} = 0.0569$	0.0164
23	1	6	$0.538(1 - \frac{1}{6}) = 0.448$	$0.0569 + \frac{1}{6\times 5} = 0.0902$	0.0181

Note how censorings reduce Y_j without producing a jump: e.g. between $t = 7$ ($Y = 17$) and $t = 10$ ($Y = 15$) the subject censored at 9^+ leaves the risk set.

Example 5.9: Nelson–Aalen estimate for the 6-MP group

t	$\tilde{H}(t) = \sum d_j/Y_j$	$\sigma_H^2 = \sum d_j/Y_j^2$	SE
$0 \le t < 6$	0	0	0
$6 \le t < 7$	$3/21 = 0.1428$	$3/21^2 = 0.0068$	0.0825
$7 \le t < 10$	$0.1428 + 1/17 = 0.2017$	$0.0068 + 1/17^2 = 0.0103$	0.1015
$10 \le t < 13$	$0.2017 + 1/15 = 0.2683$	$0.0103 + 1/15^2 = 0.0147$	0.1212
$13 \le t < 16$	$0.2683 + 1/12 = 0.3517$	$0.0147 + 1/12^2 = 0.0217$	0.1473
$16 \le t < 22$	$0.3517 + 1/11 = 0.4426$	$0.0217 + 1/11^2 = 0.0299$	0.1729
$22 \le t < 23$	$0.4426 + 1/7 = 0.5854$	$0.0299 + 1/7^2 = 0.0503$	0.2243
$23 \le t < 35$	$0.5854 + 1/6 = 0.7521$	$0.0503 + 1/6^2 = 0.0781$	0.2795

Comparison. Plotting $\tilde{S} = e^{-\tilde{H}}$ against $\hat{S}$ (K&M Fig. 4.1A), the Nelson–Aalen-based curve lies slightly *above* the KM curve at every step, and correspondingly $\tilde{H} = -\log \hat{S}$ lies slightly above $\tilde{H}$; the gap widens in the tail where Y_j is small (e.g. at $t = 23$: $\hat{S} = 0.448$ vs. $\tilde{S} = e^{-0.752} = 0.471$).

8 Selected exercises

Motivation. Three structural facts about the estimators of this lecture: without censoring they reduce to the classical empirical quantities, the Kaplan–Meier estimator redistributes each censored subject over its own future, and the Nelson–Aalen estimator matches the exact moments of the true cumulative hazard at the order statistics.

Exercise 5.1 (No censoring: KM and Greenwood reduce to binomial quantities)

Suppose no observation is censored ($\delta_i = 1$ for all i; ties allowed).
(a) Show that $\hat{S}(t) = 1 - \mathbb{F}_n(t)$ for all t.
(b) Show that Greenwood's formula reduces to $\hat{V}[\hat{S}(t)] = \hat{S}(t)\{1-\hat{S}(t)\}/n$ whenever $\hat{S}(t) > 0$.

Solution [Supplement].

(a)

Step 1. Put $Y_{D+1} = 0$. Without censoring, a subject with $T_i \ge t_j$ either dies at t_j or has $T_i \ge t_{j+1}$,

so $Y_{j+1} = Y_j - d_j$ and $Y_1 = n$.

Step 2. Hence $1 - d_j/Y_j = Y_{j+1}/Y_j$, and for $t_k \le t < t_{k+1}$ (with $t_{D+1} = \infty$) the product telescopes:

$$\hat{S}(t) = \prod_{j=1}^{k} \frac{Y_{j+1}}{Y_j} = \frac{Y_{k+1}}{n} = \frac{\#\{i : T_i > t\}}{n} = 1 - \mathbb{F}_n(t).$$

For $t < t_1$ both sides equal 1.

(b) If $\hat{S}(t) > 0$ then $Y_{k+1} > 0$, and each Greenwood term telescopes:

$$\frac{d_j}{Y_j(Y_j - d_j)} = \frac{Y_j - Y_{j+1}}{Y_j Y_{j+1}} = \frac{1}{Y_{j+1}} - \frac{1}{Y_j}, \qquad \sum_{j=1}^{k} \frac{d_j}{Y_j(Y_j - d_j)} = \frac{1}{Y_{k+1}} - \frac{1}{n}.$$

With $\hat{S} = Y_{k+1}/n$,

$$\hat{V}[\hat{S}(t)] = \frac{Y_{k+1}^2}{n^2}\Big(\frac{1}{Y_{k+1}} - \frac{1}{n}\Big) = \frac{1}{n} \cdot \frac{Y_{k+1}}{n}\Big(1 - \frac{Y_{k+1}}{n}\Big) = \frac{\hat{S}(t)\{1 - \hat{S}(t)\}}{n},$$

the usual binomial variance estimate for the proportion $\#\{T_i > t\}/n$. ■

Exercise 5.2 (Self-consistency of the Kaplan–Meier estimator)

Let t be such that $\hat{S}(t) > 0$. Show that

$$\hat{S}(t) = \frac{1}{n}\Big[\#\{i : T_i > t\} + \sum_{i:\,\delta_i = 0,\ T_i \le t} \frac{\hat{S}(t)}{\hat{S}(T_i)}\Big].$$

That is, $\hat{S}(t)$ is the empirical survival fraction once every subject censored at $T_i \le t$ is counted as alive at t with its own estimated probability $\hat{P}(X > t \mid X > T_i)$.

Solution [Supplement].

Step 1. For $s \le t$ let $R(s) = \#\{i : T_i > s\}$ and $A(s) = \sum_{i:\delta_i=0, T_i \le s} 1/\hat{S}(T_i)$; these are well defined because $\hat{S}$ is nonincreasing and $\hat{S}(s) \ge \hat{S}(t) > 0$. Put

$$\varphi(s) = \frac{R(s)}{\hat{S}(s)} + A(s).$$

The claim is $\varphi(t) = n$, multiplied through by $\hat{S}(t)/n$.

Step 2. R, $\hat{S}$ and A are right-continuous step functions that change only at observed times T_i. For $s < \min_i T_i$, $\varphi(s) = n/1 + 0 = n$.

Step 3. Let $s \le t$ be an observed time, $Y = \#\{i : T_i \ge s\} = R(s-)$, with d deaths and c censorings at s. Then $R(s) = Y - d - c$, $A(s) = A(s-) + c/\hat{S}(s)$ and $\hat{S}(s) = \hat{S}(s-)(1 - d/Y)$ (also when $d = 0$). Hence

$$\varphi(s) - \varphi(s-) = \frac{Y - d - c}{\hat{S}(s)} + \frac{c}{\hat{S}(s)} - \frac{Y}{\hat{S}(s-)} = \frac{(Y - d) - Y(1 - d/Y)}{\hat{S}(s)} = 0.$$

The tie convention matters here: the c subjects censored at s are in Y, which is what makes $\hat{S}(s) = \hat{S}(s-)(1 - d/Y)$.

Step 4. By Steps 2–3, φ is constant, equal to n, on $(-\infty, t]$, which proves the identity.

Efron (1967) showed that this "redistribute-to-the-right" property characterizes $\hat{S}$; it is the template for Turnbull's algorithms in Lecture 8. ■

Exercise 5.3 (Nelson–Aalen at the order statistics)

Let $X_1, \dots, X_n$ be i.i.d. with continuous F, no censoring, and order statistics $X_{(1)} < \cdots < X_{(n)}$.
(a) Show that $\tilde{H}(X_{(k)}) = \sum_{i=1}^{k} \frac{1}{n-i+1}$ and $\sigma_H^2(X_{(k)}) = \sum_{i=1}^{k} \frac{1}{(n-i+1)^2}$.
(b) Show that $\mathrm{E}\{H(X_{(k)})\} = \tilde{H}(X_{(k)})$ and $\mathrm{Var}\{H(X_{(k)})\} = \sigma_H^2(X_{(k)})$.

Solution [Supplement].

(a) Continuity of F excludes ties a.s., so at $t_i = X_{(i)}$ we have $d_i = 1$ and $Y_i = n - i + 1$. Substitute into Theorem 5.5.

(b)

Step 1. Since F is continuous, $S(X) = 1 - F(X) \sim U(0,1)$, so for $u \geq 0$, $P\{H(X) > u\} = P\{S(X) < e^{-u}\} = e^{-u}$: the $E_i = H(X_i)$ are i.i.d. Exp(1).

Step 2. H is nondecreasing, so it preserves order: $H(X_{(k)}) = E_{(k)}$, the k-th order statistic of $E_1, \dots, E_n$.

Step 3. The joint density of $(E_{(1)}, \dots, E_{(n)})$ is $n! \exp(-\sum_i e_i)$ on $0 < e_1 < \cdots < e_n$. Let $D_i = E_{(i)} - E_{(i-1)}$, $E_{(0)} = 0$; this linear map is triangular with unit diagonal, so its Jacobian is 1, and $\sum_i e_i = \sum_i (n-i+1) d_i$. Since $\prod_{i=1}^{n} (n-i+1) = n!$, the density of $(D_1, \dots, D_n)$ is

$$\prod_{i=1}^{n} (n-i+1)\, e^{-(n-i+1)d_i}, \qquad d_i > 0,$$

so the spacings are independent with $D_i \sim \mathrm{Exp}(n-i+1)$.

Step 4. $E_{(k)} = \sum_{i=1}^{k} D_i$, hence $\mathrm{E}E_{(k)} = \sum_{i \leq k} (n-i+1)^{-1}$ and, by independence, $\mathrm{Var}E_{(k)} = \sum_{i \leq k} (n-i+1)^{-2}$. Compare with (a).

So the Nelson–Aalen estimate at $X_{(k)}$ is exactly the mean of the true cumulative hazard there, and Aalen's variance is exactly its variance. ■